

A Short Note on Uniformly Traced Coproduct Categories

Elena Di Lavore

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Categories with coproducts and a uniform trace are a good model for the control flow of programs: coproducts give semantics to choice statements and traces give semantics to loops. This note recalls the definition of uniformly traced coproduct category and the history of the ideas leading to it.

Definition 0.1. A category, $\mathbb{C}$, with coproducts is *uniformly traced* if there is an operator, $\mathbf{tr}_S: \mathbb{C}(S + A, S + B) \rightarrow \mathbb{C}(A, B)$ for all objects S , A and B of $\mathbb{C}$, that satisfies the axioms below.

$$\begin{aligned}
 \mathbf{tr}_S((\mathbb{1}_S + u) \circledast f \circledast (\mathbb{1}_S + v)) &= u \circledast \mathbf{tr}_S(f) \circledast v && \text{(Tightening)} \\
 \mathbf{tr}_S(f + g) &= \mathbf{tr}_S(f) + g && \text{(Strength)} \\
 \mathbf{tr}_T \mathbf{tr}_S f &= \mathbf{tr}_{S+T} f && \text{(Joining)} \\
 \mathbf{tr}_0 f &= f && \text{(Vanishing)} \\
 \mathbf{tr}_A \sigma_{A,A}^+ &= \mathbb{1}_A && \text{(Yanking)} \\
 \text{if } f \circledast (u + \mathbb{1}_B) &= (u + \mathbb{1}_A) \circledast g, \text{ then } \mathbf{tr}_S f = \mathbf{tr}_T g && \text{(Uniformity)}
 \end{aligned}$$

Sliding vs Uniformity. The better known *sliding axiom* follows from uniformity: $((u + \mathbb{1}_A) \circledast h) \circledast (u + \mathbb{1}_B) = (u + \mathbb{1}_A) \circledast (h \circledast (u + \mathbb{1}_B))$ implies $\mathbf{tr}_S((u + \mathbb{1}_A) \circledast h) = \mathbf{tr}_T(h \circledast (u + \mathbb{1}_B))$.

Feedback vs Trace. Delayed trace [5, 6], or feedback [11, 12], is a generalisation of trace that drops the yanking axiom: the feedback loop is not instantaneous but creates a delay, $\partial_A = \mathbf{tr}_A(\sigma_{A,A}^+)$.

Traces and fixpoints for control-flow recursion. Trace operators were introduced for generic braided monoidal categories [10], but, when the monoidal

product is the categorical (co)product, they are equivalent to fixpoint operators [7, 14]. Fixpoint operators are an earlier axiomatisation of recursion [3, 15, 16] that builds on the work of Elgot [2, 4] and Scott [13]. The uniformity axiom appears in this work to introduce “simulation equivalence” [8, 15, 16].

Partially additive categories support a notion of iteration that generalises iteration theories and Scott semantics [1]; Kleisli categories of partially additive monads are partially additive and are uniformly traced with the coproduct [9]. Kleene monads, instead, ensure that their Kleisli category is additive and supports a Kleene star operator [5].

Example 0.2. In Kleisli categories of partially additive monads [9, Definition 4.2], the trace is constructed via coalgebras and it is automatically uniform [9, Theorem 4.2]. In particular, the countably-supported subdistribution monad is partially additive [9, Example 4.4].

An interesting consequence of the uniformity axiom is that the initial object is also terminal. The unique maps via the zero object may be interpreted as “diverge” because they are the semantics of the statement “while true do skip”.

Lemma 0.3. *In a uniformly traced cocartesian category, the initial object is also terminal.*

Proof. Define morphisms $!_A: A \rightarrow 0$ as $!_A := \mathbf{tr}_A([\mathbb{1}_A, \mathbb{1}_A])$. These assemble into a natural transformation: by the universal property of coproducts, we have

$$(f + f) \circ [\mathbb{1}_B, \mathbb{1}_B] = [f, f] = [\mathbb{1}_A, \mathbb{1}_A] \circ f$$

by uniformity, we obtain

$$f \circ !_B := f \circ \mathbf{tr}_B([\mathbb{1}_B, \mathbb{1}_B]) = \mathbf{tr}_A([\mathbb{1}_A, \mathbb{1}_A]) = !_A$$

Therefore, for every object A , there is a unique morphism $!_A: A \rightarrow 0$; this makes the object 0 terminal. $\square$

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